

Reverse Stable Diffusion 2.0

Jimmy, Jason, Woody



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Stable Diffusion - Image to Prompts Kaggle Competition

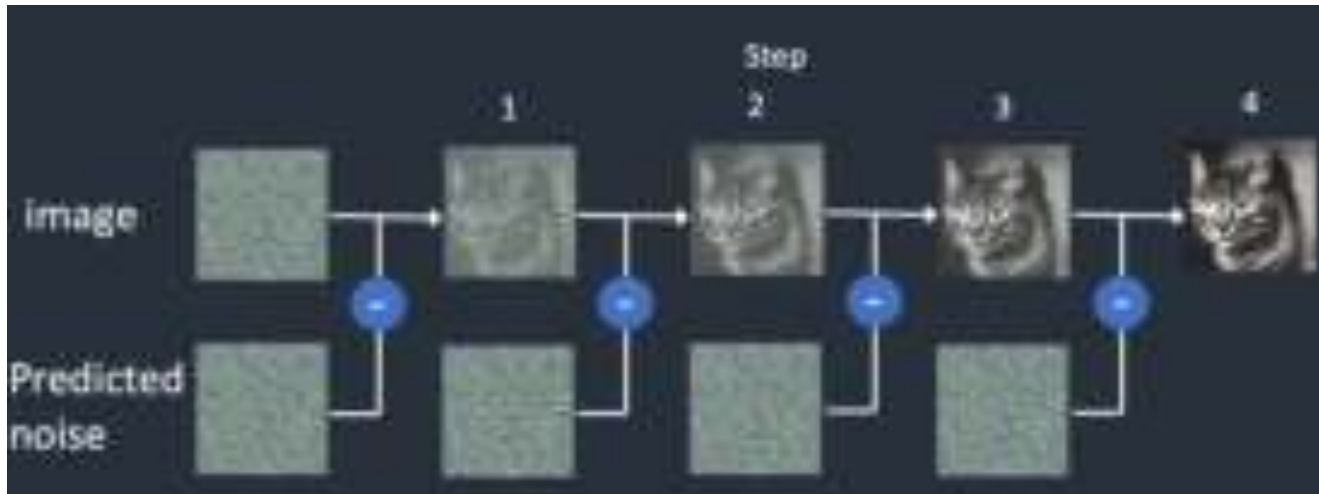
- reverse the typical direction of a generative text-to-image model
- Evaluate by comparing the Sentence Transformer Cosine Similarity of the original and generated prompt



Kaggle employee Phil at a donut shop ordering all the best donuts, with a speech bubble that proclaims "Donuts. It's what's for dinner!"

Stable Diffusion 1

- Latent text-to-image diffusion model capable of generating photo-realistic images given any text input, trained on 512x512 images from a subset of the LAION-5B database
- Trained to denoise random Gaussian noise step by step with CLIP guidance



Stable Diffusion 2

- OpenCLIP, developed with LAION-2B dataset
- Default resolutions of both 512x512 pixels and 768x768 pixels
- Trained on “aesthetic subset of the LAION-5B dataset” created by Stability AI and further filtered with NSFW filters

Stable Diffusion 2

- OpenCLIP, developed with LAION-2B dataset
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1.5: jumping cat, DSLR photo, photorealistic



2.0: jumping cat, DSLR photo, photorealistic

Dataset - DiffusionDB-2M

Filter:

- Square images, dimension of 512x512 pixels
- Prompt: five or more words, English language, without NULL, null, or NaN, and does not start or end the same as any other prompt
- removed images with prompt similarity of 90% or higher using Faiss vector search

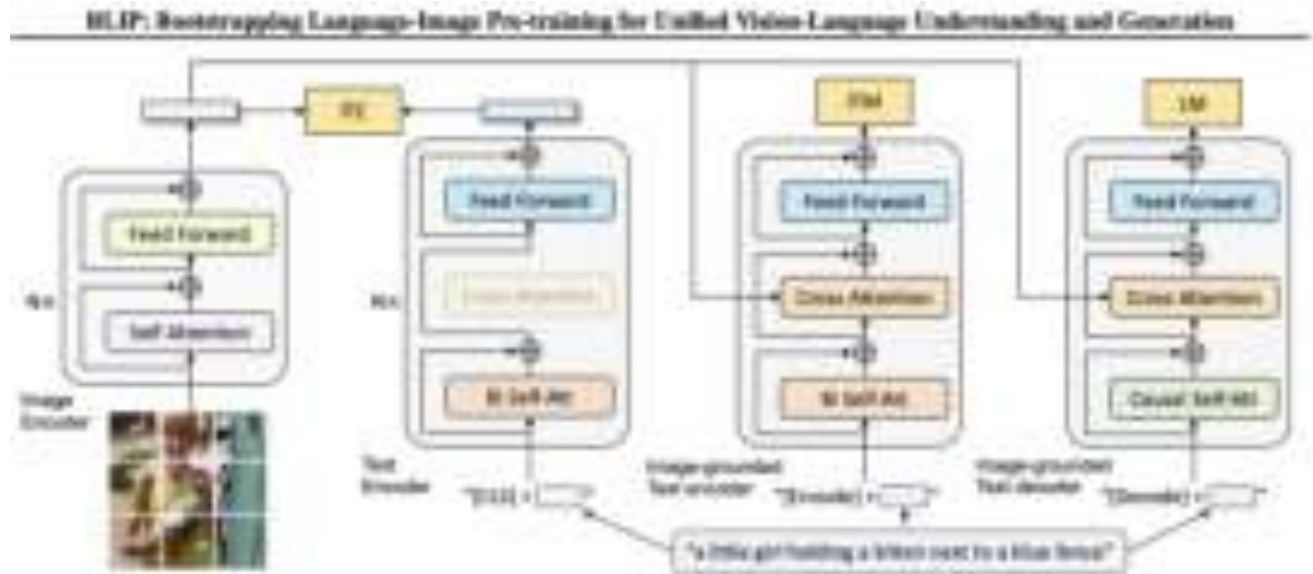
Reduce the dataset size from 2m to 150k

Methods

- BLIP
- GIT
- Segment Anything Model (SAM) GIT (SAMGIT)
- Combined/Ensemble Methods
 - CLIP Interrogator
 - SAM + BLIP
 - SAM + BLIP + CLIP Interrogator

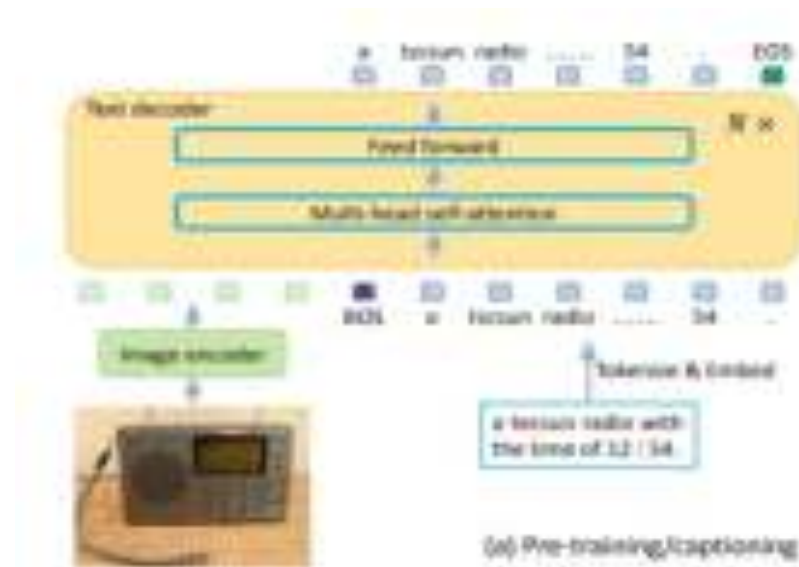
BLIP: Bootstrapping Language-Image Pre-training for Unified Vision-Language Understanding and Generation

BLIP uses ViT as its image encoder to receive input of images, and implements a Unimodal encoder, an Image-grounded text encoder, and an Image-grounded text decoder to perform various tasks



Generative Image-to-Text Base

- Simple architecture:
 - Image encoder: Use CLIP as image encoder pretrain on 10M image-text pairs or 4M images, which is a combination of COCO, SBU, CC3M and VG.
 - Text decoder: Transformer module, decoded in an autoregressive way



Segment Anything Model (SAM)

- zero-shot image segmentation
- points, bounding boxes, masks, (potentially text) as "prompts" to produce image segmentations



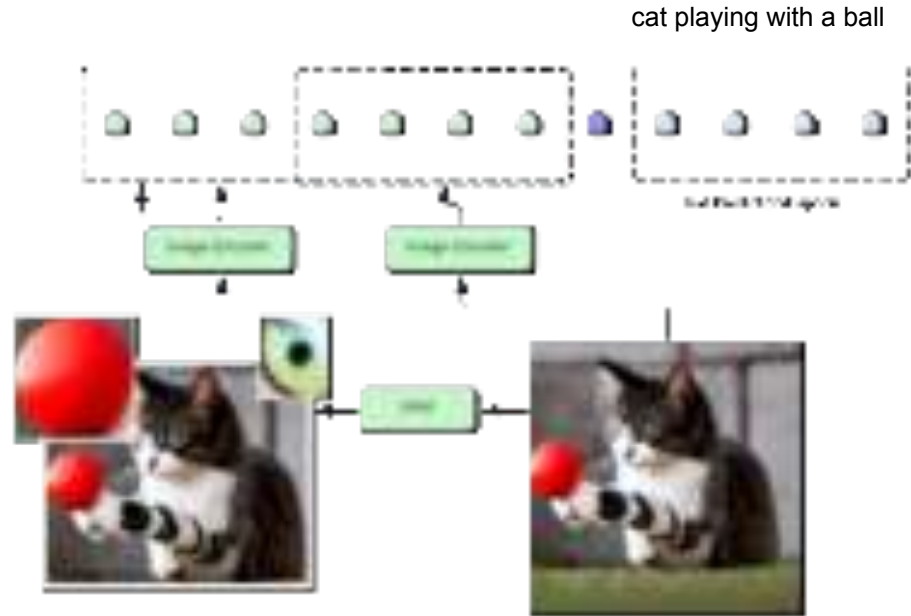
Segment Anything Model (SAM)

- Top 3 IoU



SAMGIT

- Use SAM detect and crop object features
- Feed them all into GIT



Combined/Ensemble Methods

- Combining multiple models
- We achieved this by appending the text of the methods to each other

CLIP Interrogator

- BLIP Caption
- CLIP cosine similarity to list of common words: “medium.txt” (image type), “movement.txt” (image style), “flavors.txt” (image description)



a gray and white cat playing with a red ball, a picture, photorealism, pouncing, awesome cat, robotic cat

SAM+BLIP

- Use SAM to find segments, and label with directly with BLIP
- Final output is

“There is a cat that is playing with a ball on the grass, there is a close up of a green eye with a black pupil, there is a red ball sitting on a white surface”



SAM+BLIP+CLIP Interrogator

There is a cat that is playing with a ball on the grass, there is a close up of a green eye with a black pupil, there is a red ball sitting on a white surface, a picture, photorealism, pouncing, awesome cat, robotic cat

Results

	Bleu_4	METEOR	ROUGE-L	CIDEr	SPICE	Kaggle Score	7-arg Eval
Pre-trained Baselines							
BLIP	0.243	0.253	0.478	0.848	0.199	0.401	0.373
GIT	0.411	0.302	0.602	1.329	0.236	0.371	0.353
Fine-tuned Models							
BLIP	0.050	0.121	0.285	0.172	0.087	0.346	0.298
GIT	0.104	0.154	0.325	0.370	0.102	0.390	0.361
SAMGIT	0.144	0.160	0.395	0.440	DNF	DNF	0.456
Ensemble Methods							
CLIP Int	-	-	-	-	-	0.458	0.422
SAM+BLIP	-	-	-	-	-	DNF	0.365
SAM+BLIP+CLIP	-	-	-	-	-	0.416	0.393

Table 1: The first group is pre-trained models as baselines. The second group is fine-tuned models on filtered DiffusionDB. The third group is ensemble methods using pre-trained models (no fine-tuning). DNF means that the model ran too slow in evaluation or failed due to errors. "-" means that we did not run the model/ensemble on the score.

- SAMGIT seems to have the least drift from COCO Eval
- CLIP Interrogator surprisingly has the best Kaggle Score
- We could not get SAMGIT to run fast enough, even with the lowest settings for SAM
- We do see a correlation between Kaggle Score and 7-img Eval

	Res. 4	SETHUR	BUXUE-L	CLIP	SPICE	Kaggle Score	7-img Eval
Pre-trained Baselines							
BLIP	0.245	0.215	0.476	0.600	0.196	0.464	0.373
CLIP	0.001	0.000	0.000	1.109	0.290	0.371	0.335
Fine-tuned Models							
BLIP	0.090	0.071	0.285	0.172	0.007	0.346	0.299
CLIP	0.184	0.054	0.325	0.170	0.000	0.396	0.362
SAMGIT	0.144	0.000	0.205	0.400	TIME	TIME	0.436
Ensemble Methods							
CLIP+BLIP	-	-	-	-	-	0.488	0.410
SAM+BLIP	-	-	-	-	-	TIME	0.365
SAM+BLIP+CLIP	-	-	-	-	-	0.436	0.393

Table 1: The first group is pre-trained models as baselines. The second group is fine-tuned models on different datasets. The third group is ensemble methods using pre-trained models (see fine-tuned). TIME means that the model ran too slow in evaluation or failed due to errors. "-" means that we did not test that (no results/ensemble on the score).

Conclusion

- If we could run SAMGIT fast enough, it probably would do well in Kaggle
- Issues:
 - Through observation, it doesn't seem like SAMGIT is really using the "SAM" object crop parts
 - SAM ranked by IoU isn't optimal for detecting "relevant objects"
 - Our dataset was scraped from the internet, with most of the generated images generated from SD1.5

GIT-base: the story of the dinosaur

GIT fine-tuned: cartoon blue - eyed dinosaur in forest, forest, cartoon, cartoon

SAMGIT fine-tuned: a dinosaur in the middle of a forest

BLIP-large base: cartoon dinosaur - like dinosaur - like dinosaur with a cheeseburger

BLIP-large fine-tuned: a dinosaur in 4 q in the woods, trending on artstation, central composition, very cartoon, shape focus, high detailed, by anton fadeev



ultrasaurus holding a black bean taco in the woods, near an identical cheneosaurus

CLIP Interrogator: a cartoon dinosaur with a piece of cheese in its mouth, an illustration of, sumatraism, mmmmm, buttercup eating pizza, pastry lizard

SAM+BLIP: a cartoon dinosaur with a piece of cheese in its mouth, a cartoon dinosaur holding a piece of cheese, a cartoon dinosaur holding a piece of cheese

SAM+BLIP+CLIP: a cartoon dinosaur holding a piece of cheese, a cartoon dinosaur holding a piece of cheese, an illustration of, sumatraism, mmmmm, buttercup eating pizza, pastry lizard



ultrasaurus holding a black bean taco in the woods, near an identical cheneosaurus